

PEP27 Budget — Princeton Electric Speedboating

Team: Princeton Electric Speedboating · **Craft:** Osprey · **Division:** Uncrewed — Autonomy · **Status:** Returning national team (\$5,000) **Prepared:** 22 September 2026 · **Form due:** Monday 28 September 2026

Summary

Osprey is built, sealed and proven on the water at 30 mph. We are not funding a new boat this year. Every line below buys one of three things: **testing time, the fix for a known failure, or the autonomy system the division requires.** That is where a returning team's money should go, and it is what the Risk rubric now weights most heavily.

	AMOUNT
PEP funding requested	\$5,000
Sponsor contribution requested (Castle Creations, pending)	\$1,480
Total program value	\$6,480

Budget by form line

Prices marked † are current quoted retail. Everything else is our estimate.

FORM LINE	ITEMS	EXPECTED PURCHASE	PEP	SPONSOR
Hull	Resin, carbon patch, sealant, fasteners. Reseal and float-test after winter storage; no new hull.	Oct 2026	\$300	—
Energy Storage	8 × Lectron Pro 22.2V 5200mAh 100C, EC5 †\$139.99	Nov 2026	\$1,120	—
Motor(s)	2 × Castle 2535 680Kv †\$349.95	Oct 2026	—	\$700
Control System	2 × Castle Hydra Cobra 5 HV 12S marine ESC †\$389.95	Oct 2026	—	\$780
	Autopilot, GPS, telemetry radio for the autonomy division	Nov 2026	\$450	—
	Rudder blade and mount, steering servo spare	Dec 2026	\$200	—
Propeller(s)	Test range of Graupner K-series, ~15 pitch/diameter combinations at ~\$7 each	Oct 2026	\$200	—
Other	Raw-water pumps \$80 · driveline spares — collets, flex shaft, stuffing tube, thrust bearings \$250 · wiring, QS8 connectors, fusing \$150 · flow and temperature instrumentation \$100 · shop consumables \$120	Oct–Dec 2026	\$700	—
Contingency	Held against back-order slip and test-day breakage	as needed	\$630	—
Subtotal			\$3,600	\$1,480
Travel to Event	Princeton NJ → Portsmouth VA, ~300 mi each way. Van, fuel, two nights lodging for the team	Apr 2027	\$1,400	—
Total			\$5,000	\$1,480

Why these lines

Energy storage is a testing line, not a power line. The pack set is adequate for a race. It is not adequate for a season of testing: the previous team skipped range testing entirely rather than risk draining packs to damage. Doubling the pack count turns a single run per afternoon into a full test day, which is the direct route to the in-water testing the Risk rubric now rewards.

Propellers are the cheapest speed on the boat. Osprey drew 45 A at 30 mph where the powertrain was designed for roughly 140 A. The motors and controllers are not the limit — the propeller is not loading them. A \$200 set of pitches and diameters is the highest-return line on this form.

The driveline spares are against a known failure. The flex shaft is retained in its collet by clamping force alone, and at PEP26 it worked loose mid-race. We are redesigning the retention and will need parts to iterate.

Autonomy is new spend. Osprey has never driven itself. The autopilot, GPS and telemetry are the division requirement, including the kill-on-navigation-loss the rules require us to demonstrate.

Castle Creations sponsorship

We are approaching Castle Creations for two 2535 680Kv motors and two Hydra Cobra 5 HV marine ESCs. Osprey runs their powertrain already and their Castle Link logging is what gave us the data we are working from.

Status: requested, not committed. If it does not come through we absorb the \$1,480 as follows: reduce the pack purchase from eight to four (−\$560) and release the contingency (−\$630), covering \$1,190. The remaining **\$290** comes from club funds, which we hold independently of PEP. **No line above is contingent on the sponsorship arriving** — it changes how much testing we get, not whether the boat races.

Note on the ESC. The Hydra Cobra 5 HV ships with a machined water-cooled heatsink as standard, so this is not an upgrade request — it is the same part we already run. What we are fixing this year is the raw-water side feeding it, which never flowed reliably.

Schedule risk

Both Castle parts are on back order at retail as of this week. That is the main schedule risk in this budget, and it is why motors and ESCs are dated October rather than later. Ordering early, whether sponsored or purchased, protects the winter build window. Propellers are also dated October so that prop testing can begin as soon as the boat is back in the water.

Princeton Electric Speedboating · Graham [Surname], President · graham@smithsnet.us